

Chapter 23: User interface solutions

Let's say we get inspired of some ideas from modern user interfaces used in programs and convert them to React components. That's what **"User interface solutions"** topic consists of. Attribution is appreciated.

Section 23.1: Basic Pane

```
import React from 'react';

class Pane extends React.Component {
  constructor(props) {
    super(props);
  }

  render() {
    return React.createElement(
      'section', this.props
    );
  }
}
```

Section 23.2: Panel

```
import React from 'react';

class Panel extends React.Component {
  constructor(props) {
    super(props);
  }

  render(...elements) {
    var props = Object.assign({
      className: this.props.active ? 'active' : '',
      tabIndex: -1
    }, this.props);

    var css = this.css();
    if (css !== '') {
      elements.unshift(React.createElement(
        'style', null,
        css
      ));
    }

    return React.createElement(
      'div', props,
      ...elements
    );
  }

  static title() {
    return '';
  }

  static css() {
    return '';
  }
}
```

Major differences from simple pane are:

- panel has focus in instance when it is called by script or clicked by mouse;
- panel has `title` static method per component, so it may be extended by other panel component with overridden `title` (reason here is that function can be then called again on rendering for localization purposes, but in bounds of this example `title` doesn't make sense);
- it can contain individual stylesheet declared in `css` static method (you can pre-load file contents from `PANEL.css`).

Section 23.3: Tab

```
import React from 'react';

class Tab extends React.Component {
  constructor(props) {
    super(props);
  }

  render() {
    var props = Object.assign({
      className: this.props.active ? 'active' : ''
    }, this.props);
    return React.createElement(
      'li', props,
      React.createElement(
        'span', props,
        props.panelClass.title()
      )
    );
  }
}
```

`panelClass` property of `Tab` instance must contain class of *panel* used for description.

Section 23.4: PanelGroup

```
import React from 'react';
import Tab from './Tab.js';

class PanelGroup extends React.Component {
  constructor(props) {
    super(props);
    this.setState({
      panels: props.panels
    });
  }

  render() {
    this.tabSet = [];
    this.panelSet = [];
    for (let panelData of this.state.panels) {
      var tabIsActive = this.state.activeTab == panelData.name;
      this.tabSet.push(React.createElement(
        Tab, {
          name: panelData.name,
          active: tabIsActive,
          panelClass: panelData.class,
          onMouseDown: () => this.openTab(panelData.name)
        }
      ));
    }
  }
}
```

```

    ));
    this.panelSet.push(React.createElement(
      panelData.class, {
        id: panelData.name,
        active: tabIsActive,
        ref: tabIsActive ? 'activePanel' : null
      }
    ));
  }
  return React.createElement(
    'div', { className: 'PanelGroup' },
    React.createElement(
      'nav', null,
      React.createElement(
        'ul', null,
        ...this.tabSet
      )
    ),
    ...this.panelSet
  );
}

openTab(name) {
  this.setState({ activeTab: name });
  this.findDOMNode(this.refs.activePanel).focus();
}
}

```

`panels` property of `PanelGroup` instance must contain array with objects. Every object there declares important data about panels:

- `name` - identifier of panel used by controller script;
- `class` - panel's class.

Don't forget to set property `activeTab` to name of needed tab.

Clarification

When tab is down, needed panel is getting class name `active` on DOM element (means that it gonna be visible) and it's focused now.

Section 23.5: Example view with `PanelGroup`s

```

import React from 'react';
import Pane from './components/Pane.js';
import Panel from './components/Panel.js';
import PanelGroup from './components/PanelGroup.js';

class MainView extends React.Component {
  constructor(props) {
    super(props);
  }

  render() {
    return React.createElement(
      'main', null,
      React.createElement(
        Pane, { id: 'common' },
        React.createElement(
          PanelGroup, {

```

```

        panels: [
            {
                name: 'console',
                panelClass: ConsolePanel
            },
            {
                name: 'figures',
                panelClass: FiguresPanel
            }
        ],
        activeTab: 'console'
    }
)
),
React.createElement(
    Pane, { id: 'side' },
    React.createElement(
        PanelGroup, {
            panels: [
                {
                    name: 'properties',
                    panelClass: PropertiesPanel
                }
            ],
            activeTab: 'properties'
        }
    )
)
);
}
}

```

```

class ConsolePanel extends Panel {
    constructor(props) {
        super(props);
    }

    static title() {
        return 'Console';
    }
}

```

```

class FiguresPanel extends Panel {
    constructor(props) {
        super(props);
    }

    static title() {
        return 'Figures';
    }
}

```

```

class PropertiesPanel extends Panel {
    constructor(props) {
        super(props);
    }

    static title() {
        return 'Properties';
    }
}

```